

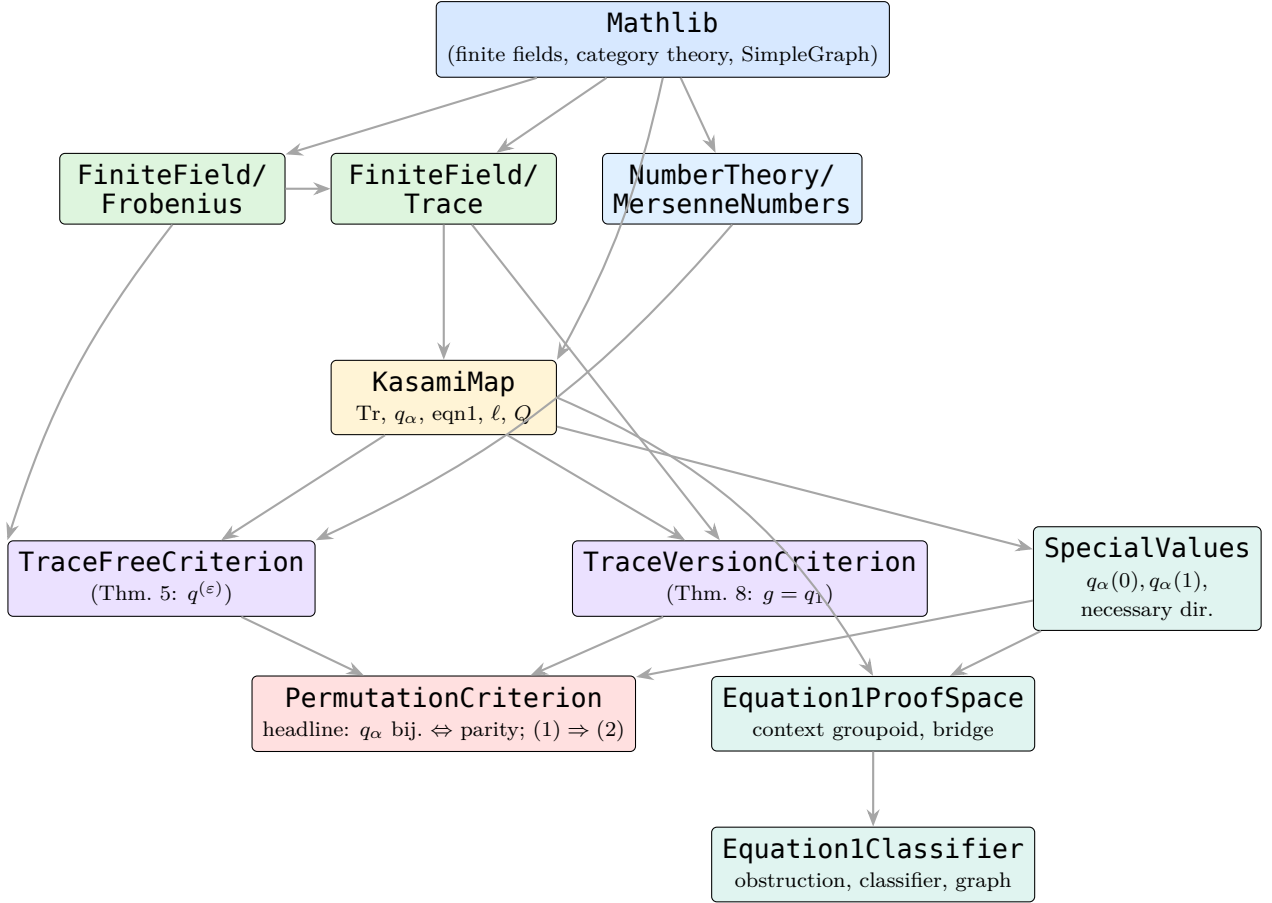
Equation (1): the simplified proof-space structure

Module DAG, the classifier equivalence, and the parity graph

A companion to `docs/Equation1_Shortcuts_and_Redundancies.md`. All declarations named here build and are `sorry`-free on the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

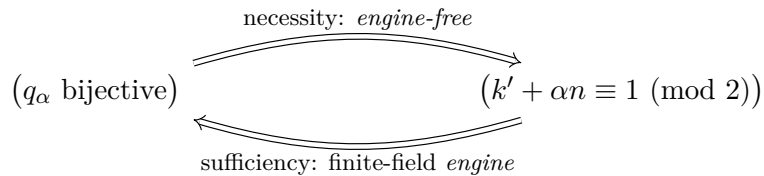
1 The module dependency DAG (new modules highlighted)

The Kasami permutation-criterion library, with the two new “proof space” modules (`Equation1ProofSpace`, `Equation1Classifier`) drawn in teal. An arrow $A \rightarrow B$ means “ B imports A ”.



2 The headline, split into engine-free and engine halves

The refactor isolates *all* the finite-field depth into the sufficiency ($\Leftarrow$) direction. The necessity ($\Rightarrow$) direction is a single, completely general fact.



Observation. The necessity arrow is `qKasami_bijective_imp_parity`: a bijection fixing 0 cannot send the distinct point 1 to 0, so $q_\alpha(1) \neq 0$; and $q_\alpha(1) = k' + \alpha n$. It uses **no field, no finiteness, no Frobenius** — only `Function.Bijective.injective`. The sufficiency arrow is the deep half, resting on Frobenius/Fermat `pow_card`, `add_pow_char_pow`, the Mersenne gcd, and `Finite.injective_iff_surjective`.

The general core of the necessity arrow, valid for any self-map of any pointed type:

$$f(0) = 0, \quad f \text{ bijective}, \quad c \neq 0 \implies f(c) \neq 0 \quad (\text{apply_ne_zero_of_bijective_fixing_zero}).$$

3 The classifier: one invariant, three equivalent views

Let `Context` bundle the parameters (n, k, k', α) with the single invariant $\text{par}(c) = k' + \alpha n \in \mathbb{Z}/2\mathbb{Z}$. Making “same parity” the hom-relation turns `Context` into a thin groupoid, and the invariant is a functor to the two-point discrete category.

$$\text{Context} \xrightarrow[\simeq]{\text{parFunctor}} \text{Discrete}(\mathbb{Z}/2\mathbb{Z})$$

`parFunctor` is full, faithful and essentially surjective, hence an equivalence.

The equivalence `contextClassifier : Context  $\simeq$  Discrete( $\mathbb{Z}/2\mathbb{Z}$ )` collapses the whole groupoid of contexts onto the two-point classifier; being a groupoid it is self-dual, `Context  $\simeq$  Contextop` (`contextSelfDual`).

The resulting two-class partition appears in three mutually equivalent guises:

$$\begin{array}{ccc} (\text{parGraph.Reachable } c \ d) & \xleftrightarrow{\text{iff_iso}} & (\text{Nonempty } (c \cong d)) \\ \uparrow \text{\scriptsize _reachable_iff} & \nearrow \text{\scriptsize iso_iff_par} & \\ (c.\text{par} = d.\text{par}) & & \end{array}$$

- **combinatorial:** equality of the parity bit;
- **categorical:** isomorphism in the groupoid `Context`;
- **graph-theoretic:** a *walk* in `parGraph` — a composite of single parity-preserving edges (proof arrows), whose transitive closure is reachability.

4 The analytic hinge

The bridge tying the abstract picture to the actual map is

$$q_\alpha(1) = 0 \iff \text{par}(c) = 0 \quad (\text{qKasami_one_eq_zero_iff_par}),$$

from which the criterion is read off the classifier’s object map (`qKasami_one_eq_zero_iff_classifier`) and the necessity direction lands in the “odd” component (`bijective_imp_classifier_one`).

$$\begin{array}{ccc} (q_\alpha(1) = 0) & \xleftrightarrow{\text{_iff_par}} & (\text{par}(c) = 0) \\ \uparrow \text{\scriptsize _iff_classifier} & \nearrow & \\ (\text{parFunctor.obj } c = \langle 0 \rangle) & & \end{array}$$